Pairi Daiza Foundation Grant

Cover Letter

Overview

- **Project title**: Acoustic Monitoring for Migratory Birds in Lithuania – Land of the Storks
- **Principal investigator**: Yi-Chin (Sunny) Tseng - University of Northern British Columbia, Canada
- **Local partners**: Tomas Pikūnas - bioreserve manager in Žuvintas Biosphere Reserve in Lithuania; and Modestas Bružas - biologist in Curonian Spit National Park in Lithuania
- **Current funding**: National Geographic Early Career Grant (EC-KOR-53792R-18)
- **Potential funding**: Pairi Daiza Foundation Grant for Biodiversity and Conservation Research

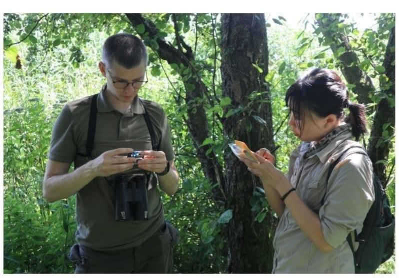
Summary

The project aims to establish Lithuania's first long-term passive acoustic monitoring program for migratory birds, providing baseline biodiversity data across key habitats. Initial pilot recordings were collected during the 2025 breeding season, demonstrating strong potential for large-scale implementation. With support from the Pairi Daiza Foundation, we will continue and expand monitoring during the 2026 breeding season at two ecologically significant sites—Žuvintas Biosphere Reserve and Curonian Spit National Park—with full collaboration from local partners.

Fieldwork will occur from May to July 2026, preceded by logistical planning and equipment preparation, and followed by data analysis and reporting by December 2026. The project will deliver a comprehensive baseline dataset of avian acoustic activity to support long-term biodiversity monitoring and conservation management in Lithuania.

Local collaborators

The project is supported by local partners in Lithuania, at Žuvintas Biosphere Reserve and Curonian Spit National Park. Principal investigator Sunny Tseng conducted pilot fieldwork with Modestas, Tomas, and Arūnas in 2025, who are committed to continuing collaboration during the 2026 season.



Curonian Spit National Park

Modestas Bružas, biologist (left)
Sunny Tseng, bioacoustic researcher (right)

Žuvintas Biosphere Reserve

Tomas Pikūnas, biosphere manager (right)
Arūnas Pranaitis, ornithologist (left)

Project Proposal

Acoustic Monitoring for Migratory Birds in Lithuania – Land of the Storks

Scientific context and primary objective

Lithuania, characterized by over 3,000 lakes, extensive mixed forests, and a rich network of wetlands, supports one of Europe's most diverse bird communities. Located along the Baltic Flyway, it provides critical stopover and breeding habitats for hundreds of migratory and resident species, from the dunes of the Curonian Spit to the wetlands of the Žuvintas Biosphere Reserve. Despite this ecological significance, systematic biodiversity monitoring using soundscapes or passive acoustic methods has not yet been established, and existing documentation of bird vocalizations is limited. Pilot acoustic recordings collected during the 2025 field season demonstrated the feasibility of using passive acoustic monitoring for migratory birds in Lithuania. This project aims to extend these efforts into a long-term program that will offer a non-invasive way to document species presence, vocal activity, and temporal patterns, providing a permanent record of avian biodiversity in Lithuania.

The primary objective of this project is to establish Lithuania's first multi-year passive acoustic monitoring program for migratory birds, creating a long-term baseline dataset to inform conservation and management. This initiative directly supports the Pairi Daiza Foundation's mission by expanding scientific knowledge, enabling habitat protection through informed management, and fostering public awareness of avian diversity.

Description of the grant project

Specific objectives

- **Characterize bird community composition and activity patterns:** Assess species composition across diverse habitats and quantify the temporal patterns of vocal activity for key species.
- **Expand and share acoustic biodiversity data:** Collect and curate both community-level soundscape recordings and species-specific vocalizations. Contribute processed datasets to open-access platforms such as Xeno-Canto and the Global Soundscape Project to support global conservation efforts.
- **Support long-term monitoring in Lithuania:** Provide national partners with data summaries and practical recommendations for integrating passive acoustic monitoring into long-term biodiversity management plan.

Study areas

Curonian Spit National Park is located between the Baltic Sea and the Curonian Lagoon. This UNESCO World Heritage Site features diverse habitats—coastal dunes, forests, and wetlands—and holds the highest recorded diversity of bird species in Lithuania. It also neighbors Ventės Ragas Ornithological Station, a key site for migratory bird research. Twelve ARU sites were established, with three recorders per habitat type (sand dune, reed, old-growth forest, and planted pine forest), each at least 3 km apart to ensure spatial independence (Fig.1).

Žuvintas Biosphere Reserve, in southern Lithuania, is the country's largest wetland complex and a UNESCO Biosphere Reserve. It serves as a breeding and staging area for more than 240 bird species, including several globally threatened taxa such as the Aquatic Warbler (*Acrocephalus paludicola*) and Great Snipe (*Gallinago media*). A total of 12 ARU sites were established, including 2 sites near the lake, 4 in marsh habitats, and 6 in forest habitats. Each ARU was located at least 1 km from the others to ensure spatial independence (Fig.1).

Data processing and analysis

Audio recordings will be processed using BirdNET, a machine-learning algorithm that automatically identifies bird species from sound. The results will yield site-specific species lists, vocal activity indices, and temporal patterns. Cleaned recordings of individual species will be manually verified and shared on Xeno-Canto, expanding Lithuania's representation in this global repository.

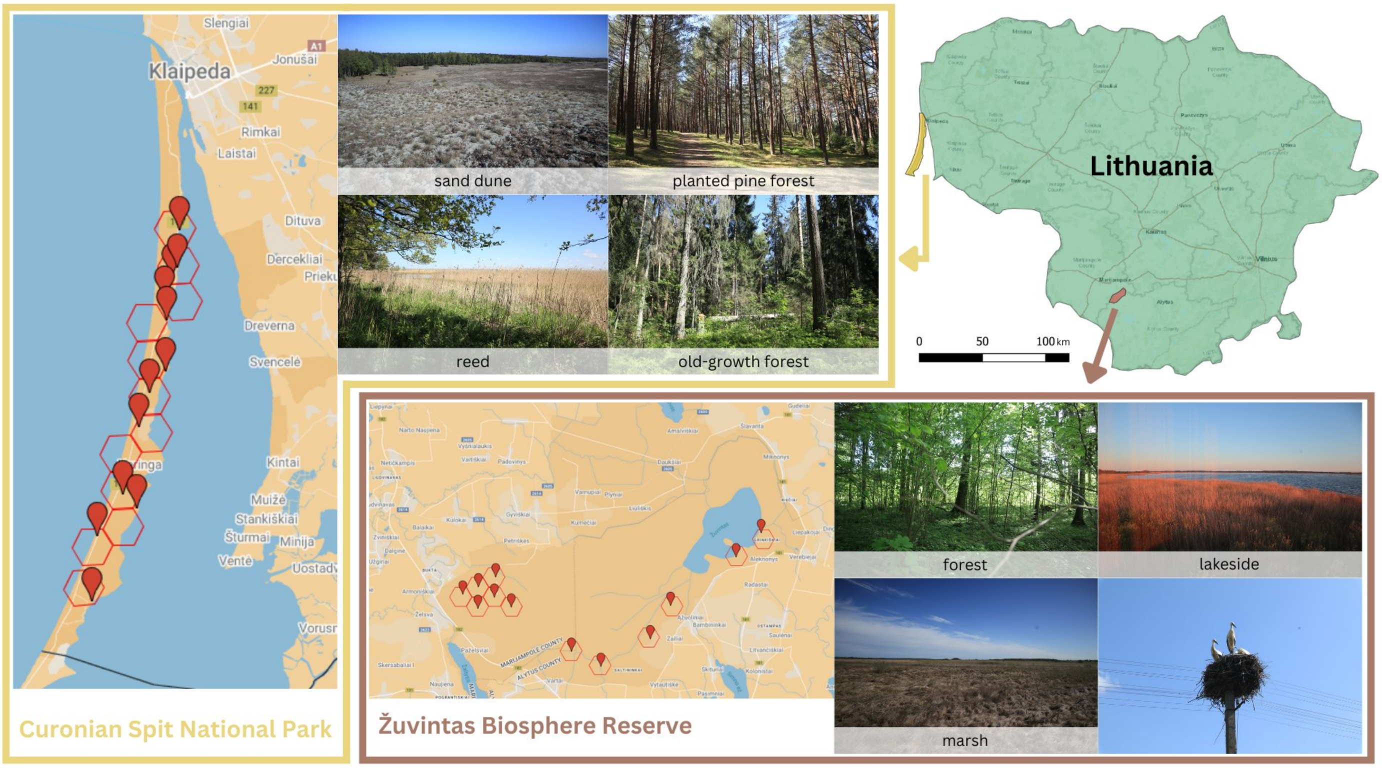


Figure 1. Study sites at Curonian Spit National Park and Žuvintas Biosphere Reserve, showing the monitored habitats: sand dune, planted pine forest, reed, and old-growth forest at Curonian Spit, and forest, lakeside, and marsh at Žuvintas. (Photo credit: Sunny Tseng)

Work plan

The early months of the project are devoted to securing permits with local collaborators, coordinating logistics, and preparing equipment. The main field season runs from May to July, followed by systematic data processing, species verification, and preliminary analyses. The detailed timeline is outlined below:

Period	Activities	Expected outcomes
Jan–Apr 2026	Confirm research permits, coordinate logistics with local partners, calibrate ARUs	All permissions and logistics finalized
May–Jul 2026	Install and monitor ARUs; conduct targeted recordings at both sites	Successful collection of acoustic data
Aug–Oct 2026	Classify recordings using BirdNET; manual verification of key species	Preliminary biodiversity summary
Nov–Dec 2026	Share data with local collaborators, upload species recordings to Xeno-Canto, prepare final report	Final dataset and report submitted to collaborators

Provisional budget

Item	Cost (EUR)	Justification
Flight	800	Round-trip transportation to Lithuania
Car rental	700	Transportation within Lithuania for 20 days (10 for deployment, 10 for retrieval)
Accommodation	700	Lodging during two 10-day field trips
Fieldwork meals	500	Food during two 10-day field trips
Data analysis	100	Software access (e.g., Raven) for advanced acoustic analysis

Total: **2,800 EUR**
Total Requested: **2,500 EUR**

Ethical aspect deadling with sample collection or use

This project is fully non-invasive and poses no risk to birds or their habitats. Acoustic monitoring is conducted passively using autonomous recorders placed along established trails or research zones with prior permission from park authorities. No handling, trapping, or disturbance of wildlife will occur.

Outreach and communication

The project has gained international recognition, with strong support from the Taiwanese Representative Office in Lithuania during the 2025 pilot fieldwork (Fig.2). This collaboration highlights the role of international partnerships in advancing biodiversity research and fostering cross-cultural exchange. To enhance the visibility and impact of the project's findings, outreach efforts will engage multiple audiences through diverse communication channels:

- **General audience (English and Mandarin)**: Building on the project's initial support from National Geographic, findings will be shared through a popular science article on National Geographic's platforms. A Mandarin version will be submitted to National Geographic Mandarin Edition, broadening access to readers in Taiwan and other Mandarin-speaking regions.
- **Lithuanian ornithologists and naturalists**: Sunny Tseng will return to Lithuania in 2026 to conduct fieldwork and host in-person presentations and workshops. Co-organized with the Taiwanese Representative Office, these events will engage the Lithuanian Ornithological Society (LOD) and universities such as Vilnius University and Vytautas Magnus University. The sessions will share project results, demonstrate acoustic monitoring techniques, and promote collaborations for long-term ecological research in Lithuania.

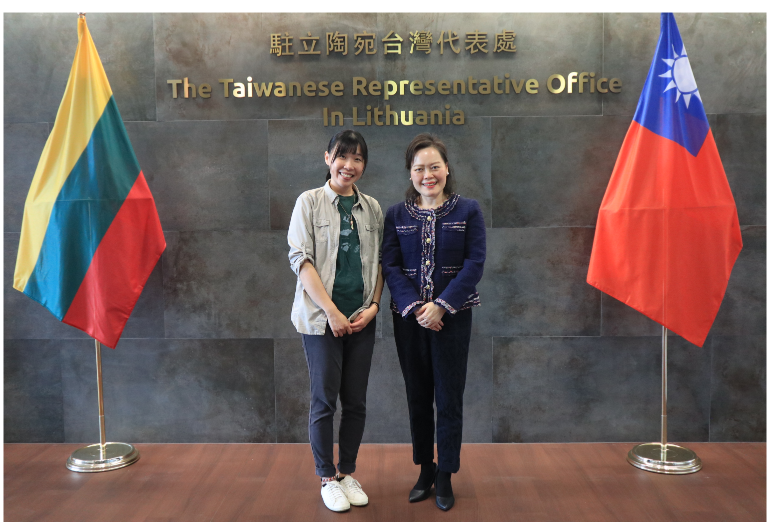


Figure 2. The principal investigator, Sunny Tseng (left), meeting with the Ambassador of Taiwan in Lithuania, Constance Hsueh-hong Wang (right), during the 2025 field season in Lithuania.

Through these outreach activities, the project will extend its impact beyond academia, inspiring both the public and scientific communities. By connecting researchers, conservation practitioners, and the general public across countries, it will foster a shared appreciation for biodiversity and the innovative tools used to protect it.

YI-CHIN TSENG
(SUNNY)

sunnytseng.ca

EDUCATION

- **Ph.D., University of Northern British Columbia** — **Prince George, Canada**
  Natural Resources & Environmental Studies — *Sep.2021 – present*
  Supervisor: Ken Otter

- **Master of Science, University of British Columbia** — **Vancouver, Canada**
  Department of Forest Resources Management — *Sep.2017 – Nov.2019*
  Supervisor: Bianca N.I. Eskelson

- **Bachelor of Science, National Taiwan University** — **Taipei, Taiwan**
  School of Forestry and Resource Conservation & Department of Physics — *Sep.2012 – Jun.2017*
  Double Major Student
  Supervisor: Biing T. Guan, Tzeng-Yih Lam, and Shi-Wei Chu

PUBLICATION

JOURNAL ARTICLE

- **Tseng, S.** and S. Kahl. **2025** birdnetTools: An R package for working with BirdNET output. *Journal of Open Source Software*. (Submitted)
- Baum, J.K., Slein, M.A., Garen, J.C., . . . , **Tseng, S.**, et al. **2025** Widespread ecological responses and cascading effects of the record-breaking 2021 North American heatwave. *Nature Ecology and Evolution*. (In Revision)
- Darras, K., Rountree, R., Van Wilgenburg, S., . . . , **Tseng, S.**, et al. **2025** Worldwide Soundscapes: a synthesis of passive acoustic monitoring across realms. *Global Ecology and Biogeography* (link)
- **Tseng, S.**, Hodder, D., and Otter, K. **2025** Setting BirdNET confidence thresholds: Species-specific vs. universal approaches. *Journal of Ornithology* (link)
- **Tseng, S.**, Hodder, D., and Otter, K. **2024** Using Autonomous Recording Units for Vocal Individuality: Insights from Barred Owl Identification. *Avian Conservation and Ecology* (link)
- Pelletier, F., Eskelson, B.N.I., Monleon, V.J., and **Tseng, Y.-C. 2021** Using Landsat imagery to assess burn severity of National Forest Inventory plots. *Remote Sensing* (link)
- **Tseng, Y.-C.**, Eskelson, B.N.I., Martin, K., and LeMay, V. **2021**. Automatic bird sound detection: Logistic model based acoustic occupancy model. *Bioacoustics* (link)
- **Tseng, Y.-C.**, and Chu, S.-W. **2017**. High spatio-temporal-resolution detection of chlorophyll fluorescence dynamics from a single chloroplast with confocal imaging fluorometer. *Plant Methods* (link)

OPEN-SOURCE SOFTWARE PACKAGES

- **birdnetTools** (2025) — R package for post-processing, visualization, and validation of BirdNET output. Developed in collaboration with the BirdNET team. Available on GitHub. Documentation
- **bbsTaiwan** (2024) — Tools for accessing and analyzing Taiwan Breeding Bird Survey (BBS) data. Developed for national biodiversity research and public engagement. Documentation

POPULAR SCIENCE ARTICLE

- **2024** Why do we band birds at all? *Nature Conservation Quarterly*, 124. (link)
- **2023** Nest boxes for barn owl conservation. *Nature Conservation Quarterly*, 122. (link)
- **2023** When beaver dam collapses. *Nature Conservation Quarterly*, 121. (link)
- **2023** Listening to the whispers of the wings. *Discovery*, 50(1):85-90. (link)
- **2022** Hey, there are wild black bears in my backyard. *Nature Conservation Quarterly*, 117. (link)

CONFERENCE TALK / POSTER (check full list here)

- **SCO-SOC Annual Meeting** — **Saskatoon, Canada**
  BirdNET workshop organizer, invited seminar speaker — *Aug.2025*
- **Ecological Data Science Summit** — **Santa Barbara, USA**
  Invited by conference organizer for discussion of current AI use in ecology — *Feb.2025*
- **Posit::conf** — **Seattle, USA**
  Selected as one of the 40 opportunity scholars, early career software developer — *Aug.2024*
- **AFO SCO-SOC WOS Joint Meeting** — **Peoria, USA**
  Evaluation and application of BirdNET: identification of bird species by their sounds (Oral) — *Aug.2024*
- **International Ecoacoustic Congress** — **Madrid, Spain**
  Setting BirdNET confidence thresholds: Species-specific vs. universal approaches (Oral) — *Jul.2024*
- **Congress of Animal Behavior and Ecology** — **Taipei, Taiwan**
  Avian biodiversity across wide spatiotemporal scale using passive acoustic monitoring (Oral) — *Feb.2024*

RECOGNITION

SCHOLARSHIP and RESEARCH GRANT (check full list here)

- **Bert Brink scholarship** — **BC Nature**
  Outstanding graduate student for natural resources research in BC — *Aug.2025*
- **Michael Smith Foreign Study Supplements** — **NSERC**
  Developing an R package for BirdNET in Chemnitz, Germany — *June.2025*
- **Early Career Grant** — **National Geographic**
  Leading a fieldwork in Lithuania for avian acoustic monitoring — *May.2025*
- **Hugh Hamilton Memorial Scholarship** — **Nature Vancouver**
  For graduate student with outstanding academic & involvement with Nature Science. — *May.2024*
- **rOpenSci Championship Program** — **rOpenSci**
  Outstanding proposal for R package development. — *Jan.2024*
- **Canada Graduate Scholarship - Doctorial (CGS-D)** — **NSERC**
  Rewarding and retaining high-calibre doctoral students at Canadian institutions. — *Apr.2023*

AWARD (check full list here)

- **SCO-SOC Taverner Award** — **SCO-SOC**
  Awarded for outstanding ornithological research in Canada — *Apr.2025*
- **Student Presentation Award** — **SCO-SOC**
  Best oral presentation in the annual conference in Peoria, Illinois — *Aug.2024*
- **Best Student Presentation Award** — **BC & CAN TWS**
  Best student oral presentation in The Wildlife Society conference — *Mar.2023*